SECTION 15

STEEL REINFORCEMENT

GLOSSARY OF TERMS

Reinforcement connector 15.01 Reinforcement connector is a coupler or sleeve designed to transmit the force between two bars in tension or compression.

Reinforcement 15.02 Reinforcement is steel bar, wire or fabric used in its untensioned state in concrete.

MATERIALS

Reinforcing steels 15.03 Except as stated in Clause 15.04 below, reinforcing steels for the reinforcement shall comply with the following:

Category	Reinforcement	Compliance
A	Steel reinforcing bars covered by CS2 Amd 1/2025	CS2
B	Other reinforcing steel bars Grade 500 not covered by CS2 Amd 1/2025	BS 4449
C	Plain round steel bars (diameter > 12 mm) of Grade 250 not covered by CS2 Amd 1/2025	BS 4449:1997
D	Plain round steel bars (diameter ≤ 12 mm) of Grade 250 not covered by CS2 Amd 1/2025	BS 4482
E	Steel fabric sheets Amd 1/2025	BS 4483

Stainless steel reinforcement 15.04 Stainless steel bars for the reinforcement shall be ribbed bar to BS 6744. The steel designation numbers are in accordance with BS EN 10088-1. The requirements on sampling, testing and acceptance criteria shall be in accordance with BS 6744 except otherwise stated.

Epoxy coatings to reinforcement 15.05 (1) Epoxy coatings to reinforcement and patching materials for epoxy coatings shall comply with BS ISO 14654 except as stated in Clauses 15.05(2), 15.22, 15.32, 15.34 and 15.38. The coatings shall be applied by the electrostatic spray method complying with BS ISO 14654 at a factory approved by the Engineer.

(2) The film thickness of the coating after curing shall be at least 0.17 mm and shall not exceed 0.30 mm over the complete periphery including deformations and ribs. The bond classification of coated bars determined in bond performance tests shall not be less than that of uncoated bars.

Hot dip galvanizing to reinforcement 15.06 (1) Hot dip galvanizing to reinforcement shall comply with BS EN ISO 1461. The galvanization shall be applied after cutting and bending of the reinforcement.

(2) Materials for repair to hot dip galvanized reinforcement shall comply with BS EN ISO 1461.

Reinforcement connectors 15.07 (1) Reinforcement connectors shall be of a proprietary type approved by the Engineer.

(2) Reinforcement connectors for tension joints shall be a cold swaged or threaded type. The connectors shall be capable of developing the full tensile strength of the parent bar and shall comprise high tensile steel studs and seamless steel tubes fitted with protective plastic caps.

(3) Reinforcement connectors for compression joints shall be of a wedge locking or bolted sleeve type.

Cover spacers 15.08 (1) Cover spacers for reinforcement shall be concrete blocks or of a proprietary plastic or concrete type. Proprietary plastic and concrete cover spacers shall be of a type approved by the Engineer.

(2) Cover spacers for Class F3, F4 and F5 finishes shall be of a proprietary plastic or concrete type. Cover spacers for epoxy-coated reinforcement and galvanized reinforcement shall be of a proprietary plastic type.

(3) Cover spacers shall be as small as practicable consistent with their purpose and shall be designed to maintain the specified cover to reinforcement. Cover spacers shall be capable of supporting the weight of reinforcement and construction loads without breaking, deforming or overturning.

(4) The strength and durability of concrete blocks and proprietary concrete cover spacers shall not be less than that of the surrounding concrete.

(5) Cover spacers for Class F3, F4 and F5 finishes shall be of a colour similar to that of the surrounding concrete and shall not cause indentations in the formwork.

Chairs, supports and spacers 15.09 Chairs, supports and spacers other than cover spacers for reinforcement shall be steel. The steel shall be coated with nylon, epoxy, plastic or other dielectric material for epoxy-coated reinforcement and shall be hot dip galvanized for galvanized reinforcement.

Tying wire 15.10 Tying wire for reinforcement adjacent to and above Class F4 and F5 finishes and stainless steel reinforcement shall be 1.2 mm diameter stainless steel wire. Tying wire for epoxy-coated reinforcement shall be 1.6 mm diameter soft annealed steel wire coated with nylon, epoxy, plastic or other dielectric material. Tying wire for galvanized reinforcement shall be 1.6 mm diameter galvanized soft annealed steel wire. Tying wire for other reinforcement shall be 1.6 mm diameter soft annealed steel wire.

Tying devices and clips 15.11 Tying devices and clips for reinforcement shall be of a proprietary steel type approved by the Engineer. Tying devices and clips for reinforcement adjacent to and above Class F4 and F5 finishes shall be stainless steel. Tying devices and clips for epoxy-coated reinforcement shall be coated with nylon, epoxy, plastic or other dielectric material. Tying devices and clips for galvanized reinforcement shall be galvanized.

SUBMISSIONS

Particulars of reinforcement

15.12 (1) The following particulars of the proposed steel reinforcement shall be submitted to the Engineer for approval Amd 3/2021:

(a) For Category A (i.e. CS2) reinforcement, a certificate from the quality assured stockist and a copy of the manufacturer's certificate/document in accordance with CS2 Cl. 4.1.

(b) For other reinforcement, a certificate/document from the manufacturer showing the manufacturer's name, the place of manufacture and showing that the reinforcement complies with the requirements stated in Clause 15.03, including:
- Chemical composition (cast analysis) and calculation of carbon equivalent value
- Dimensions and mass per metre of bar/wire Amd 1/2025
- Tensile properties and bend performance
- Bond property or evidence that the bond property has complied with the requirements
- Sheet dimensions and shear force of welded joints for steel fabric sheet

(2) The particulars together with other requirements stated in the Contract shall be submitted to the Engineer for approval Amd 3/2021 for each batch of reinforcement delivered to the Site and at least 14 days before fixing of the reinforcement starts.

Particulars of epoxy coatings to reinforcement

15.13 (1) The following particulars of the proposed epoxy coatings to reinforcement shall be submitted to the Engineer for approval Amd 3/2021:

(a) Name and location of the coating factory,

(b) Date and place of the coating application, and

(c) Original Certificate(s) of the coating materials in compliance with BS ISO 14656, including:
- Corrosion resistance
- Chemical resistance
- Cathodic disbondment
- Salt spray resistance
- Abrasion resistance
- Impact strength
- Coating flexibility

The above tests shall be carried out once every 5 years or when there are changes in the composition of the coating materials whichever is the earlier.

(2) The particulars shall be submitted to the Engineer for approval Amd 3/2021 at least 14 days before the first delivery of epoxy-coated reinforcement to the Site. Certificates together with the particulars of the reinforcement shall be submitted to the Engineer for approval Amd 3/2021 for each batch of epoxy-coated reinforcement delivered to the Site and at least 14 days before fixing of the reinforcement starts.

Particulars of galvanized coatings to reinforcement

15.14 (1) The following particulars of the proposed galvanized coatings to reinforcement shall be submitted to the Engineer for approval Amd 3/2021:

(a) Name and location of the coating factory, and

(b) Original certificate from the manufacturer showing the date and place of application of the coating and showing that the galvanized coatings comply with the requirements stated in the Contract and including results of tests carried out by methods as recommended in BS EN ISO 1461 for thickness of coating.

(2) The particulars shall be submitted to the Engineer for approval Amd 3/2021 at least 14 days before the first delivery of galvanized reinforcement to the Site. Certificates together with the particulars of the reinforcement shall be submitted to the Engineer for approval Amd 3/2021 for each batch of galvanized reinforcement delivered to the Site and at least 14 days before fixing of the reinforcement starts.

Particulars of reinforcement connectors

15.15 Particulars of the proposed materials and methods of installation for reinforcement connectors, including the manufacturer's literature, shall be submitted to the Engineer for approval Amd 3/2021 at least 28 days before fixing of reinforcement connectors starts.

Bending schedules

15.16 Bending schedules of reinforcement complying with BS 8666 shall be prepared by the Contractor and submitted to the Engineer for information Amd 3/2021 before bending of reinforcement starts.

Samples of materials

15.17 Samples of the following proposed materials shall be submitted to the Engineer for information Amd 3/2021 at the same time as particulars of the materials are submitted:

(a) Reinforcement,

(b) Epoxy-coated reinforcement,

(c) Galvanized reinforcement,

(d) Reinforcement connectors for tension joints and compression joints,

(e) Cover spacers and other spacers, and

(f) Tying wire, tying devices and clips.

HANDLING AND STORAGE OF MATERIALS

Handling of reinforcement

15.18 (1) Reinforcement shall not be subjected to rough handling, shock loading or dropping from a height.

(2) Nylon, rope or padded slings shall be used for lifting epoxy-coated reinforcement and galvanized reinforcement. Bundles shall be lifted with a strong back or with multiple supports to prevent abrasion.

Storage of

15.19 (1) Reinforcement shall be stored off a levelled, well drained and

reinforcement

maintained hard-standing ground on level supports and in a manner, which will not result in damage or deformation to the reinforcement, or in contamination of the reinforcement. Measures to protect the reinforcement from wetting and rusting caused by weather shall be proposed for the Engineer's approval. Reinforcement shall be stored horizontally.

(2) Different types and sizes of reinforcement shall be stored separately.

(3) Reinforcement shall not be stored on or adjacent to concrete surfaces that form part of the permanent work.

(4) Epoxy-coated reinforcement and galvanized reinforcement shall be stored on wooden or padded cribbing.

CUTTING AND BENDING REINFORCEMENT

Cutting and bending reinforcement

15.20 (1) Reinforcement shall be cut and bent in accordance with BS 8666 to the specified shapes and dimensions and shall be bent at temperatures of at least 5°C and not exceeding 100°C.

(2) Epoxy tends to become brittle at lower temperatures, additional care should be exercised when fabrication is performed during cold weather. Surfaces of bending equipment in contact with epoxy-coated reinforcement shall be fitted with nylon or plastic mandrels.

(3) Grade 500 and stainless steel reinforcement shall not be rebent or straightened after bending. Grade 250 reinforcement which projects from the hardened concrete may be bent aside and rebent provided that the internal radius of the bend is at least twice the diameter of the bar and that bending is not carried out by levering against the concrete or by other methods which in the opinion of the Engineer are likely to damage the concrete.

(4) The ends of bars to be used with reinforcement connectors for compression joints shall be sawn square with all burrs removed.

SURFACE CONDITION OF REINFORCEMENT

Surface condition of reinforcement

15.21 (1) Reinforcement shall be clean at the time of fixing and shall be free of loose mill scale, loose rust or any substance that, in the opinion of the Engineer, is likely to reduce the bond or affect the reinforcement or concrete chemically. The reinforcement shall be maintained in this condition until concrete is placed around it.

(2) If the surface condition of the reinforcement deteriorates such that it does not comply with the requirements stated in Clause 15.21(1), the reinforcement shall be cleaned or dealt with by other methods agreed by the Engineer.

Repairs to epoxy coatings and galvanized coatings

15.22 (1) If the coating to epoxy-coated reinforcement is delaminated or split at any point or if the coating to epoxy-coated reinforcement or galvanized reinforcement is damaged,

(a) at any point by an amount exceeding 25 mm^2 in area or 50 mm in length, or

(b) at more than three points in a 1 m length by amounts each even not exceeding 25 mm^2 in area or 50 mm in length,

that part of the reinforcement shall not be used in the permanent works. If the coating to epoxy-coated reinforcement or galvanized reinforcement is damaged at more than six points in the cut and bent length of a bar by amounts each even not exceeding 25 mm^2 in area or 50 mm in length, that length of bar shall not be used in the permanent work.

(2) Except as stated in Clause 15.22(1), all damaged areas and cut ends of epoxy-coated reinforcement shall be repaired using sealing material applied in accordance with the manufacturer's recommendations. The sealing material shall comply with BS ISO 14654 and shall be submitted to the Engineer for approval.

(3) Except as stated in Clause 15.22(1), all damaged areas and cut ends of galvanized reinforcement shall be repaired by materials to be approved by the Engineer. Sufficient material shall be applied to provide a coating of at least the same thickness as the galvanized coating. The Contractor shall refer to Section 6.3 and Annex C of BS EN ISO 1461 for advice on repair of damaged areas.

(4) Repairs to epoxy coatings and galvanized coatings shall be carried out within 8 hours of cutting or damage. Traces of rust shall be removed from the surface of the reinforcement before the repair is carried out.

FIXING REINFORCEMENT AND REINFORCEMENT CONNECTORS Amd 1/2025

Fixing reinforcement 15.23 (1) Reinforcement and reinforcement connectors for tension joints from each batch shall not be fixed until testing of the batch has been completed.

(2) Reinforcement shall be fixed rigidly in position and secured against displacement.

(3) A sufficient number of intersecting and lapping bars shall be tied using tying wire, tying devices or clips to prevent movement of the reinforcement. The ends of tying wire, tying devices and clips shall not encroach into the cover to reinforcement.

(4) Laps and joints in reinforcement shall be made only at the specified positions and by the specified method.

(5) Sufficient numbers of cover spacers, chairs, supports and spacers other than cover spacers shall be provided to maintain the reinforcement in the correct location and to maintain the specified cover at all positions. Cover spacers, chairs, supports and spacers other than cover spacers shall be placed at a maximum spacing of 1.5 m. Chairs, supports and spacers other than cover spacers shall be positioned adjacent to or above cover spacers and shall have at least the same cover as that specified for the reinforcement.

(6) Prefabricated reinforcement cages shall be adequately supported and braced before lifting.

(7) Reinforcement which is free-standing shall be secured in position and braced to prevent movement due to wind and other loads.

(8) For treated finishes, all reinforcement shall be positioned in such a way that the specified concrete cover can be achieved after the applied treatment.

Fixing reinforcement connectors 15.24 Reinforcement connectors shall be fixed in accordance with the manufacturer's recommendations and using equipment recommended by the manufacturer.

Welding of reinforcement 15.25 Reinforcement shall not be welded unless approved by the Engineer.

Exposed reinforcement 15.26 Reinforcement that is to be left exposed shall be protected by coating with cement slurry or by other methods agreed by the Engineer. Reinforcement shall be re-coated by the same method if in the opinion of the Engineer the original coating has begun to deteriorate.

Access over reinforcement 15.27 Reinforcement shall not be contaminated or displaced as a result of access over the reinforcement. Access shall be obtained by using planks and ladders or by other methods agreed by the Engineer.

TOLERANCES

Tolerances: reinforcement 15.28 (1) Tolerances on cutting and bending reinforcement shall comply with BS 8666 Table 5.

(2) The actual concrete cover to the outermost layer of reinforcement (including links, stirrups, surface reinforcement, etc.) shall be within 5 mm tolerance of the cover specified on the Drawings.

INSPECTION OF REINFORCEMENT

Inspection of reinforcement 15.29 The Contractor shall allow the Engineer to inspect the completed reinforcement before carrying out any work, including erecting formwork adjacent to reinforcement, which will make access to the reinforcement difficult. The Contractor shall inform the Engineer 24 hours, or such shorter period agreed by the Engineer, before carrying out such work.

TESTING: REINFORCEMENT AND REINFORCEMENT CONNECTORS Amd 1/2025

Batch: reinforcement and reinforcement connectors Amd 1/2025 15.30 (1) For the purpose of testing, the Category A (i.e. CS2) reinforcement delivered to the Site is to be subdivided into batches in accordance with CS2.

(2) A batch of other reinforcement or reinforcement connectors is any quantity of reinforcement or reinforcement connectors of the same type, size

and grade, manufactured by the same mill, covered by the same mill and testing certificates and delivered to the Site at any one time. In addition, for epoxy-coated reinforcement and galvanized reinforcement, the coatings shall have been applied at the same coating factory and shall be covered by the same original test certificates with original signatures and official authorization chop.

Samples: reinforcement and reinforcement connectors Amd 1/2025

15.31 (1) Samples of reinforcement and reinforcement connectors for tension joints, except for epoxy-coated reinforcement, shall be provided from each batch of the material delivered to the Site and at least 14 days before fixing of the reinforcement starts. For epoxy-coated reinforcement, samples shall be provided at least 20 working days before fixing of the reinforcement starts. The number of samples to be provided from each batch shall be as stated in Table 15.1.

(2) The number of specimens in each sample shall be as follows:

Description	Number of specimens
Category A reinforcement (i.e. steel reinforcing bars covered by CS2) Amd 1/2025	In accordance with CS2 Table 10
Category B reinforcement (i.e. steel reinforcing bars of Grade 500 not covered by CS2) Amd 1/2025	In accordance with BS 4449 Cl. 8.1.2
Category C reinforcement (i.e. plain round steel bars of Grade 250 with d > 12 mm not covered by CS2) Amd 1/2025	In accordance with BS 4449:1997 Cl. 10
Category D reinforcement (i.e. plain round steel bars of Grade 250 with d ≤ 12 mm not covered by CS2) Amd 1/2025	In accordance with BS 4482 Cl. 8.1.2
Category E reinforcement (i.e. steel fabric sheets) Amd 1/2025	3 Amd 1/2025
Epoxy-coated or galvanized reinforcement	2 additional specimens
Reinforcement connectors for tension joints	3

(3) Each specimen of bar or wire reinforcement shall be at least 1 m long. Each specimen of fabric reinforcement shall be at least 1.2 m long by 1.2 m wide and shall contain at least three bars/wires in each direction. Each specimen of reinforcement connectors shall consist of one reinforcement connector joined to two lengths of bar each at least 500 mm long. The bars shall be of the same type, size and grade as the bars to which the reinforcement connector will be fixed in the permanent work.

(4) Each specimen of reinforcement shall be taken from different units in the batch. The ends of specimens shall be cut square before delivery to the laboratory.

(5) For epoxy-coated bar reinforcement, each specimen for testing epoxy coating properties shall be a 2 m length piece cut at least 1 m from the ends of a 12 m length bar. Specimens shall be selected from different bundles of the

reinforcement batch.

Table 15.1: Rate of sampling of reinforcement and reinforcement connectors Amd 1/2025

Description	Size of batch	No. of samples per batch
Reinforcement (except steel fabric sheets) Amd 1/2025	All sizes	1
Steel fabric sheets Amd 1/2025	0 – 50 tonnes Amd 1/2025	1 Amd 1/2025
	Each additional 50 tonnes or part of 50 tonnes Amd 1/2025	1 Amd 1/2025
Reinforcement connectors for tension joints	less than 100 no.	1
	100 - 500 no.	2
	exceeding 500 no.	3

Testing: reinforcement and reinforcement connectors Amd 1/2025

15.32 (1) Each sample of reinforcement shall be tested in accordance with the following requirements:

Description	Testing requirement
Category A reinforcement (i.e. steel reinforcing bars covered by CS2) Amd 1/2025	CS2 Cl. 5.1.1
Category B reinforcement (i.e. steel reinforcing bars of Grade 500 not covered by CS2) Amd 1/2025	BS 4449 Cl. 8.1.2 with methods in Cl. 9
Category C reinforcement (i.e. plain round steel bars of Grade 250 with d > 12 mm not covered by CS2) Amd 1/2025	BS 4449:1997 Cl. 10
Category D reinforcement (i.e. plain round steel bars of Grade 250 with d ≤ 12 mm not covered by CS2) Amd 1/2025	BS 4482 Cl. 8.1.2 with methods in Cl. 9

(2) Each sample of steel fabric sheets (Category E) shall be tested in accordance with BS 4483 Cl. 8.1.2 with methods in Cl. 9 with following modifications:-

(a) The number of tests shall be as stated in Table 15.2; and

(b) The mass shall be measured as mass of fabric per square metre.

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(3) Each additional specimen as required in Clause 15.31(2) of epoxy-coated reinforcement shall be tested to determine the thickness, adhesion and continuity of the coating in accordance with Clause 15.33.

(4) Each additional specimen as required in Clause 15.31(2) of galvanized reinforcement shall be tested to determine the thickness of coating in accordance with BS EN ISO 1461 for compliance.

(5) Each sample of reinforcement connectors for tension joints shall be tested to determine the tensile properties in accordance with Clause 15.35.

(6) The number of tests on each sample shall be as stated in Table 15.2.

Testing: epoxy coating 15.33

(1) The thickness test shall be in accordance with Methods 7A, 7B and 7C of BS EN ISO 2808. For bars/wires of 12 mm diameter or below, only instruments which operate on magnetic flux principle (Methods 7B and 7C) can be used. For bars of 16 mm diameter or above, instruments which operate on either magnetic flux principle (Methods 7B and 7C) or magnetic pull-off principle (Method 7A) can be used. All measuring instruments shall be calibrated to an accuracy of ±5%.

(2) For thickness test, five recorded measurements shall be obtained approximately evenly spaced along each side of the test specimen (a total of ten recorded measurements per bar). A single recorded thickness measurement is the average of three individual reading obtained in between the ribs of three consecutive deformations.

(3) The adhesion and continuity tests shall be in accordance with BS ISO 14654.

Table 15.2: Number of tests on each sample of reinforcement and reinforcement connectors Amd 1/2025

Description	**Type and number of tests**								
	Tensile properties	Bend performance	Mass	Chemical composition (product analysis)	Bond property	Weld shear	Pitch dimension	Thickness, adhesion and continuity	Thickness and uniformity
Reinforcement (except steel fabric sheets)	In accordance with Clause 15.32(1)					-	-	-	-
Steel fabric sheets - sheet - longitudinal bar/wire - transverse bar/wire	- 3 3	- 1# 1#	3 - -	- - -	- - -	2# - -	- 1# 1#	- - -	- - -
Epoxy coating	-	-	-	-	-	-	-	2	-
Galvanized coating	-	-	-	-	-	-	-	-	2
Reinforcement connectors for tension joints	3	-	-	-	-	-	-	-	-

Note #: Tests to be carried out on one of the three specimens collected for Category E reinforcement as per Clause 15.31(2)

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Compliance criteria: reinforcement

15.33A (1) For Category A (i.e. CS2) reinforcement, if the results of the tests performed on the test specimens meet the requirements specified in CS2 Cl. 5.1.2, the batch shall be deemed to comply with the requirements.

(2) For Category B (i.e. Grade 500) reinforcement, if the results of the tests performed on the test specimens meet the requirements specified in BS 4449 Cl. 8.1.3.1 and 8.1.3.2 (for bond property, relative rib area only, other surface geometry parameters are optional.), the batch shall be deemed to comply with the requirements.

(3) For Category C (i.e. plain round steel bars, Grade 250 & d > 12 mm) reinforcement, if the results of the tests performed on the test specimens meet the requirements specified in BS 4449:1997 Cl. 5, 6, 8.2 and 11, the batch shall be deemed to comply with the requirements.

(4) For Category D (i.e. plain round steel bars, Grade 250 & d ≤ 12 mm) reinforcement, if the results of the tests performed on the test specimens meet the requirements specified in BS 4482 Cl. 8.1.3.1 and 8.1.3.2, the batch shall be deemed to comply with the requirements.

(5) For Category E (i.e. steel fabric sheets) reinforcement, the batch shall be deemed to comply with the requirements if the results of the tests performed on the test specimens meet the requirements as stated in Table 15.3. Amd 1/2025

Table 15.3: Specified test requirement / tolerance on steel fabric sheet

Type of Tests	Requirement / Tolerance
Pitch dimension	BS 4483 Cl. 7.3.2
Tensile properties	BS 4483 Cl. 8.1.3.1
Bend performance	BS 4483 Cl. 8.1.3.2
Weld shear	BS 4483 Cl. 8.1.3.2
Mass per square metre of steel fabric sheet with diameter of bars / wires in either direction greater than 8mm	± 4.5% of nominal mass
Mass per square metre of steel fabric sheet with diameter of bars / wires in both directions less than or equal to 8 mm	± 6.0% of nominal mass

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Compliance criteria: epoxy coating to reinforcement

15.34 The results of tests for thickness, adhesion and continuity of epoxy coatings to reinforcement shall comply with all the following requirements:

(a) Coating thickness
At least 90% of all recorded thickness measurements of coating shall be within the specified range. Thickness measurements below 0.13 mm shall be considered cause for rejection.

(b) Coating adhesion:
The adhesion test shall comply with the requirements in BS ISO 14654. The surface of the bent test piece shall not exhibit cracking, disbonding or ductile tearing when viewed under well-lit conditions using normal or corrected vision.

(c) Coating continuity

The continuity test shall comply with the requirements in BS ISO 14654. The continuity of the coating shall be free of holes, voids, cracks and damaged areas discernible to a person with normal or corrected vision.

Compliance criteria: reinforcement connectors for tension joints

15.35 The results of tensile properties tests on specimens of reinforcement connectors for tension joints shall comply with the following requirements:

(a) The tensile strength shall exceed 287.5 MPa for grade 250, 540 MPa for grade 500B and 575 MPa for grade 500C reinforcement.

(b) When a test is made of a representative gauge length assembly comprising reinforcement of the size, grade and profile to be used and a reinforcement connector for tension joints of the precise type to be used, the permanent elongation after loading to 0.6 times of the specified characteristic strength and unloading shall not exceed 0.1 mm. The gauge length shall span over the reinforcement connector.

Non-compliance: reinforcement

15.36 For Category A (i.e. CS2) reinforcement, if the result of any test fails, the acceptance of retests shall be considered in accordance with CS2 Cl. 5.1.4.

15.37 For other reinforcement not covered by CS2, if the result of any test fails, two additional test specimens shall be taken from different reinforcement of the same batch and be subjected to the test or tests which the original specimen failed. For properties other than chemical composition (product analysis), the batch shall be considered as not complying with the specified requirements for the properties if the result of any additional test fails. For chemical composition (product analysis), the batch shall be considered as not complying with the specified requirements of the property if the result of any additional test fails unless it is demonstrated by alternative method as accepted in CS2 Cl. 6.3. If all of the additional test specimens pass the retests, the batch shall be deemed to comply with the requirements. Otherwise the batch of reinforcement shall be rejected and removed from the Site.

Non-compliance: epoxy coatings

15.38 For each test specimen fails to meet the coating thickness, coating adhesion or coating continuity requirements, two additional test specimens shall be taken from the same batch and be subjected to the test or tests which the original specimen failed. If all of the additional test specimens pass the retests, the batch from which they have been taken shall be deemed to comply with the specified requirements for the properties. Otherwise this batch shall be rejected and removed from the Site.

Non-compliance: galvanized coatings

15.39 (1) If the result of any test for thickness of galvanized coatings to reinforcement does not comply with the specified requirements for the property, additional samples shall be provided from the same batch and additional tests for the property shall be carried out. The number of additional samples shall be as stated in Table 15.1.

(2) The number of specimens in each additional sample shall be as follows:

(i)	Galvanized bar/wire reinforcement	4
(ii)	Galvanized fabric reinforcement	2
(iii)	Galvanized reinforcement connectors for tension joints	4

(3) The number of tests on each additional sample shall be four.

(4) The batch shall be considered as not complying with the specified requirements for the property if the result of any additional test does not comply with the specified requirements for the property.

Non-compliance: reinforcement connectors

15.40 (1) If the result of any test for tensile properties of reinforcement connectors for tension joints does not comply with the specified requirements as stated in Clause 15.35, additional samples shall be provided from the same batch and additional tests as stated in Clause 15.35 shall be carried out. The number of additional samples shall be as stated in Table 15.1.

(2) The number of specimens in each additional sample shall be six.

(3) The number of tests on each additional sample shall be six.

(4) The batch shall be considered as not complying with the specified requirements for tensile properties if the result of any additional test does not comply with the specified requirements as stated in Clause 15.35.